

E-Commerce Return & Refund Analytics

Project Documentation

Sales Performance | Returns | Seller Risk | Product Risk | Refund Losses

Project Type	Data Analytics & Business Intelligence
Dataset	100,000 e-commerce transactions
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI
Final Output	Interactive Power BI dashboard

Executive summary: The analysis covers 100,000 transactions and identifies an overall return rate of 7.85%, ₹3.92 Cr in refunds, and ₹46.45 Cr in net revenue from ₹50.37 Cr in total revenue. Seller S010 and the Accessories category have the highest observed return rate at 26.15%.

1. Business Problem

E-commerce businesses process large numbers of orders, but returns and refunds can reduce revenue and profitability. This project analyzes transaction and return data to understand where returns occur, why customers return products, which sellers and products show higher return rates, and how refunds affect financial performance.

2. Project Objectives

- Analyze overall sales, orders, returns, refunds, and net revenue.
- Calculate the overall return rate and refund-loss percentage.
- Identify the most common return reasons.
- Evaluate seller-level return performance and identify high-risk sellers.
- Compare return rates across product categories and products.
- Analyze customer segments using available age, gender, and membership fields.
- Analyze monthly revenue and return-rate trends.
- Build an interactive Power BI dashboard for business users.
- Translate analytical results into practical business recommendations.

3. Tools & Technologies

Area	Tools
Programming & Analysis	Python, Pandas, NumPy
Visualization	Matplotlib, Seaborn
Business Intelligence	Microsoft Power BI
Data	CSV

4. Dataset

The project uses 100,000 e-commerce transaction records. Key fields include Order_ID, Order_Date, Customer_ID, Age, Gender, City, State, Membership, Product_ID, Product, Category, Brand, Seller_ID, Seller_Name, Seller_Rating, Quantity, Unit_Price, Revenue, Returned, Return_Reason, Refund_Percentage, and Refund_Amount. Additional analytical fields such as Age_Group and Month were created during the analysis.

5. Analysis Workflow

Stage	Work Performed
1. Data preparation	Reviewed structure, data types, dates, and analytical fields.
2. Overall KPIs	Calculated orders, returns, return rate, revenue, refunds, and net revenue.
3. Return analysis	Analyzed return reasons and return counts.
4. Seller analysis	Compared orders, returns, return rates, refunds, and net revenue by seller.
5. Category analysis	Compared return rates and financial impact across categories.
6. Product analysis	Identified products with high return rates and refund losses.
7. Customer analysis	Reviewed age groups, gender, and membership segments.
8. Time analysis	Reviewed monthly orders, revenue, refunds, return rates, and net revenue.
9. Power BI	Built an interactive dashboard for business-facing reporting.

6. Overall Business Performance

Metric	Result
Total Orders	100,000
Returned Orders	7,854
Overall Return Rate	7.85%
Total Revenue	₹50.37 Cr
Total Refund Amount	₹3.92 Cr
Net Revenue	₹46.45 Cr
Refund Loss Percentage	7.79%

All monetary values in this document are presented in Indian Rupees (₹ Crore) for consistency with the Power BI dashboard.

7. Return Reason Analysis

Return Reason	Returns	Percentage
Size Issue	1,143	14.55%
Wrong Item	1,141	14.53%
Damaged Product	1,136	14.46%
Quality Issue	1,124	14.31%
Changed Mind	1,109	14.12%
Late Delivery	1,102	14.03%
Defective Product	1,099	13.99%

Finding: Size Issue is the most common individual return reason, with 1,143 returns (14.55%). The seven return reasons are relatively balanced, so return reduction should address multiple drivers.

8. Seller Analysis

Seller	Total Orders	Returned	Return Rate	Refund Loss (₹ Cr)
S010	5,326	1,393	26.15%	0.63
S004	7,902	1,528	19.34%	0.96
S009	15,644	1,613	10.31%	0.38

Finding: S010 has the highest return rate at 26.15%. S004 has a lower return rate than S010 but the highest refund loss among these three high-return sellers at approximately ₹0.96 Cr.

9. Category Analysis

Category	Return Rate
Accessories	26.15%
Clothing	11.11%
Beauty	10.31%
Home & Kitchen	7.44%
Sports	6.06%
Electronics	3.76%
Books	2.30%

Finding: Accessories has the highest category return rate at 26.15%, significantly above the overall rate.

10. Product Analysis

Highest-risk product: Laptop Backpack (P004), in Accessories. It has a 26.15% return rate and approximately ₹0.63 Cr in refund loss. Potential investigation areas include product dimensions, specifications, images, descriptions, quality, packaging, and fulfillment. These are investigation areas rather than confirmed causes from the dataset.

11. Customer Analysis

Customer analysis covered the available age-group, gender, and membership fields. The purpose was to compare return behavior across customer segments and provide a basis for customer-experience and retention decisions.

12. Time Analysis

Highest monthly return rate: August 2025 — 8.46%.

Lowest monthly return rate: January 2025 — 7.24%.

Highest revenue month shown: March 2024 — ₹2.22 Cr.

Monthly return rates remained generally within the 7-8% range rather than showing a dramatic sustained spike.

13. Financial Impact

Total revenue was ₹50.37 Cr. Refunds reduced this by ₹3.92 Cr, resulting in ₹46.45 Cr in net revenue. Refunds represented approximately 7.79% of total revenue.

14. Power BI Dashboard

The final Power BI dashboard presents the analysis in a business-facing format. It contains KPI cards for Total Revenue, Net Revenue, Return Rate, Total Refund, and Returned Orders. The main visuals cover Returns by Reason, Return Rate by Seller, Return Rate by Category, Monthly Return Rate Trend, and a Seller Return Rate Ranking table.

- The seller table is sorted by Return Rate so higher-risk sellers appear first.
- The dashboard is designed for quick scanning of business performance and return risk.
- Detailed business interpretation is kept in the project documentation rather than overcrowding the dashboard.

15. Key Business Insights

- Overall return rate is 7.85%, with 7,854 returned orders out of 100,000 total orders.
- Refunds total approximately ₹3.92 Cr against ₹50.37 Cr in revenue.
- Seller S010 has the highest return rate at 26.15%.
- Accessories has the highest category return rate at 26.15%.
- Laptop Backpack is the highest-return product identified in the product analysis.
- Size Issue is the most common individual return reason at 14.55% of returned orders.
- Monthly return rates are relatively stable rather than showing a sustained large spike.

16. Business Recommendations

- Monitor high-risk sellers: Regularly review sellers with return rates materially above the business average.
- Investigate Accessories: Review product information, quality, fulfillment, and customer expectations.
- Review Laptop Backpack: Examine dimensions, specifications, descriptions, images, quality, and packaging.
- Reduce size-related returns: Improve size information, dimensions, and pre-purchase guidance.
- Track return rate and refund loss together: Return frequency alone does not capture financial impact.
- Use the Power BI dashboard operationally: Refresh and review the dashboard regularly to identify emerging risk.

17. Conclusion

This project demonstrates an end-to-end data analytics workflow from transaction data to business intelligence. Using Python, the analysis quantified sales, returns, refund losses, seller risk, category risk, product risk, customer segments, and monthly trends. Power BI then converted the results into an interactive dashboard. The strongest areas requiring further investigation are Seller S010, the Accessories category, and the Laptop Backpack, while Size Issue is the most common individual return reason.

18. Skills Demonstrated

- Python, Pandas, and NumPy for data preparation and analysis.
- Data cleaning, transformation, aggregation, merging, and feature engineering.
- Matplotlib and Seaborn for exploratory visualization.
- Power BI and DAX measures for business reporting.
- KPI design, dashboard layout, sorting, filtering, and visual formatting.
- Seller, category, product, customer, return-reason, and time-series analysis.
- Business insight generation and recommendations.